

Lesson 34: Limits & the Derivative — Practice Worksheet

Name: _____ Date: _____

Attempt every question. Show your working. The sections increase in difficulty.

Section 1 — Warm-up (foundation)

1. Find the gradient between (1, 3) and (4, 12).

2. If $f'(3) = 0$, what is happening at $x = 3$?

3. If $f'(x) < 0$, is the curve rising or falling?

4. Another notation for $f'(x)$?

5. Gradient of a horizontal line?

6. If $f'(2) = 4$, what is the gradient at $x = 2$?

Section 2 — Core practice

7. $f(x) = 2x$. Find $f'(5)$.

8. $f(x) = 3x - 1$. Find $f'(4)$.

9. $f(x) = 6x$. Find the gradient at $x = -2$.

10. $f(x) = 2x - 8$. Find where $f'(x) = 0$.

11. $f(x) = 4 - 2x$. Find $f'(1)$.

12. $f'(x) = x$. At which x is the gradient 7?

Section 3 — Challenge & exam-style

13. A curve has $f'(x) = 3x - 12$. Find where the gradient is zero and state what kind of point that may be.

14. $f'(x) = 2x$. State where the curve is increasing.

15. Interpret $f'(2) = 5$ for the curve $y = f(x)$.

16. $f'(x) = 6 - 2x$. For which x is the curve decreasing?